Estimating equations

Score
$\psi(W;\theta,\eta)$ with target θ and nuisance $\eta \in T$

Moment map
$M(\theta,\eta) = \mathbb{E}_P[\psi(W;\theta,\eta)]$
identification $M(\theta_0,\eta_0) = 0$,
$J_0 = \partial_\theta M(\theta_0,\eta_0)$ non-singular

condition on the derivative in η at the truth

Robustness for a nuisance pair (g,α)

Mixed bias property
$M(\theta_0,g,\alpha) = \mathbb{E}[S(W)(g-g_0)(\alpha-\alpha_0)]$
AIPW: $S=-1$, $M = -\langle g-g_0, \alpha-\alpha_0\rangle$
Rotnitzky et al. 2021

implies

Double robustness
$M(\theta_0,g,\alpha_0) = M(\theta_0,g_0,\alpha) = 0$
for all g and α

implies for a product nuisance set and linear D_0

converse if $M(\theta_0,g_0,\cdot)=0$ and $M(\theta_0,\cdot,\alpha_0)$ is affine (Chernozhukov et al. 2022, Thm. 4)

Orthogonal without double robustness
$M_{\mathrm{PO}} = \langle \ell-\ell_0, m-m_0\rangle - \theta_0\|m-m_0\|^2$
for ψ_{PO} with $\theta_0 \neq 0$

bounded S gives a bias of order $\|\hat g - g_0\|\,\|\hat\alpha-\alpha_0\|$

First-order Neyman orthogonality

Neyman orthogonality
$\mathrm{D}_0[\eta-\eta_0] = \partial_r M\big(\theta_0, \eta_0 + r(\eta-\eta_0)\big)\big|_{r=0} = 0$
for all $\eta \in \mathcal{T}_N$
near-orthogonality $\|\mathrm{D}_0[\eta-\eta_0]\| \le \lambda_N = o(N^{-1/2})$
Chernozhukov et al. 2018, Defs. 2.1–2.2

equivalent under conditions

Equivalent formulations
conditional derivatives vanish given X (Mackey et al.)
local robustness along paths of laws (Chernozhukov et al. 2022)
no first-step adjustment (Newey 1994)
$\psi_0 \perp$ nuisance tangent space (Tsiatis 2006)
orthogonal losses (Foster and Syrgkanis 2023)
s_θ uncorrelated with s_η
(Neyman 1959; Cox and Reid 1987)

Construction
$\nu(W;g) - \theta + A(Z)\{Y-g(Z)\}$ is orthogonal in g
exactly when $A = \alpha_0$, the Riesz representer
partially linear model: instruments with $\mathbb{E}[A \mid X] = 0$
parametric model: residual score $s_\theta - I_{\theta\eta} I_{\eta\eta}^{-1} s_\eta$ (Neyman)

higher derivatives

Higher-order orthogonality

Orthogonality of order q
derivatives of $\eta \mapsto M(\theta_0,\eta)$
at η_0 vanish up to order q
Mackey et al. 2018

allows

Slower nuisance rates suffice
$o_P(n^{-1/(2q+2)})$ in $L^{2(q+1)}$,
and $q=1$ gives $o_P(n^{-1/4})$

Orthogonal scores
$\psi_{\mathrm{DR}} = \nu(W;g) - \theta + \alpha(Z)\{Y-g(Z)\}$ (AIPW)
ψ_{IV} and ψ_{PO} in the partially linear model

estimating equation

Limits and alternatives
no 2-orthogonal score with $J_0 \neq 0$ in the partially linear model with Gaussian treatment noise (Mackey et al.)
any order in parametric moment models (Bonhomme et al. 2026)
higher-order influence functions (Robins et al. 2008, 2017)
product rate minimax optimal without further structure (Balakrishnan et al. 2026; Jin and Syrgkanis 2024)

Estimation and inference

DML estimator with cross-fitting
$\frac{1}{n}\sum_k \sum_{i\in I_k} \psi(W_i;\theta,\hat\eta_k) = 0$
score average at $\theta_0 = T_1 + T_2 + T_3$,
controlled by the CLT (T_1), cross-fitting (T_2),
orthogonality (first order of T_3) and rates (second order of T_3)

Rate double robustness
$\sqrt{n}(\hat\theta-\theta_0)$ asymptotically normal when both nuisance estimators are consistent and $\|\hat g - g_0\|\,\|\hat\alpha-\alpha_0\| = o_P(n^{-1/2})$
Smucler et al. 2019

product rate in place of $n^{-1/4}$ for each nuisance

Asymptotic normality
$\sqrt{n}(\hat\theta-\theta_0) = -J_0^{-1}\mathbb{G}_n\psi_0 + o_P(1) \rightsquigarrow N(0,\sigma^2)$
if $\|\hat\eta_k - \eta_0\| = o_P(n^{-1/4})$ for smooth scores
Chernozhukov et al. 2018, Thm. 3.1

analogue without a score, regularity along outcome shifts

Limits of orthogonality
removes only the first-order part of T_3; needs overlap for $\alpha_0 \in L^2$; identification assumptions are separate; defined only for estimators with a score

influence function $-J_0^{-1}\psi_0$

Semiparametric efficiency

Efficient influence function φ
unique gradient of θ in a locally nonparametric model
$-J_0^{-1}\psi_0 = \varphi$ when ψ identifies θ along the model

Orthogonality without a correction term
targeting and TMLE, $\mathbb{P}_n[\hat\alpha\{Y-\hat g\}] = 0$
sieve plug-ins and balancing weights
priors invariant along the least-favourable direction
(semiparametric Bernstein–von Mises)
debiased PFNs with label or prior correction
Melnychuk et al. 2026

reach the same influence function

Semiparametric efficiency bound
$\mathrm{Var}\,\varphi$, the smallest asymptotic variance of regular estimators, attained by DML when $-J_0^{-1}\psi_0 = \varphi$
ψ_{PO} attains it only if $\mathrm{Var}(\varepsilon \mid D, X)$ is constant

last limit

Estimators without a score

First-order insensitivity (proposal)
shift $Y \mapsto Y + \delta h(Z)$, bias $b_n(\delta;h)$
$\partial_\delta b_n(0;h) = 0$

expected imbalance

Implicit weights $w_i = n\,\partial\hat\theta/\partial Y_i$ (proposal)
$\partial_\delta b_n(0;h) = \mathbb{E}\big[\frac{1}{n}\sum_i w_i h(Z_i)\big] - \langle \alpha_0, h\rangle$
zero when the weights balance h as α_0 does